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Even End Semester Examination June, 2024

Name of the Course: **B. Tech Biotechnology** Semester: **6th**

Name of the Paper: **Animal Biotechnology** Paper Code: **TBT-601**

Time: Three Hours Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

Q1	(2x10 = 20 marks)	
(a)	**Describe method of cell counting by hemocytometer along with the derivation of its formula.**	CO-1
(b)	**Differentiate between primary cells and a cell line. Give methods of primary cell isolation in detail.**	
(c)	**What is cryopreservation? Describe the mechanism of cryopreservation with diagrams.**	
Q2	(2x10 = 20 marks)	
(a)	**What is DNA microinjection? Give details of steps for any transgenic animal created by this method.**	CO-2
(b)	**Differentiate between YAC and BAC vectors for a transgenic animal production.**	
(c)	**Write short Notes on: a) SCNT b) IVF**	
Q3	(2x10 = 20 marks)	
(a)	**What is genome mapping? Describe methods of gene mapping in detail.**	CO-3
(b)	**What is the mechanism of gene therapy? Describe the Luxturna™ gene therapy in detail.**	
(c)	**What are chemical carcinogens? Give their examples and method of testing a chemical carcinogen in laboratory.**	
Q4	(2x10 = 20 marks)	
(a)	**What is RNA antisense technology? Describe its mechanism with a diagram.**	CO-4
(b)	**Describe different methods of cell line immortalization.**	
(c)	**Discuss an example of gene cloning in a mammalian cell line.**	
Q5	(2x10 = 20 marks)	
(a)	**What is marine biotechnology? Describe its various application for human welfare.**	CO-5
(b)	**Describe the strategies for transgenic fish production. Give various applications where fish bioengineering is useful.**	
(c)	**Write short notes on: a) Ornamental fishes b) Contamination & antibiotics.**	

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TBT–602

B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

PLANT BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are Plant Growth Regulators ? Explain its type with suitable example. (CO1)

   (b) Explain the direct and indirect way of plant tissue culture. Write the advantages and limitations of PTC. (CO1)

   (c) What is somaclonal variation ? Explain the molecular basis of somaclonal variation. (CO1)

2. (a) What is haploid plant production ? Discuss the advantages and limitations of Androgenesis and gynogenesis. (CO2)

   (b) Explain the method of protoplast isolation and fusion. Write advantages and limitations also. (CO2)

P. T. O.

(c) What is synthetic seed ? Explain the method of artificial seed production. (CO2)

3. (a) Discuss the role of Forest Biotechnology in conservation of threatened plant species. (CO3)

(b) Write short notes on the following : (CO3)

(i) Cryopreservation

(ii) Green revolution

(c) What is Nitrogen fixation ? Explain different methods of Nitrogen fixation. (CO3)

4. (a) What is Genetic transformation ? Explain the methods of genetic transformation. (CO4)

(b) Explain the role of Agrobacterium tumefacience in genetic transformation. (CO4)

(c) Write short notes on the following : (CO4)

(i) T-DNA

(ii) Agrobacterium rhizogenes

5. (a) What are Secondary metabolites ? How Plant tissue culture technique can be helpful for enhanced production of secondary metabolites. (CO5)

(b) Discuss social and legal issues associated with transgenic plants. (CO5)

(c) Write short notes on the following : (CO5)

(i) CRISPR-Cas

(ii) Secondary metabolites

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TBT–603

B. TECH. BIOTECHNOLOGY (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2024

BIOSEPARATION TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the unit operations involved in downstream processing. (CO1)

   (b) Explain the production and downstream processing steps in citric acid production. (CO1)

   (c) Discuss 5 types of the products obtained in Bioprocess Industry. (CO1)

2. (a) What are various methods of cell disruption ? Explain in detail. (CO2)

   (b) Explain principle of filtration. Discuss about Rotary vacuum continuous filters. (CO2)

   (c) What is principle of Solvent extraction ? Explain any one type of centrifuge. (CO2)

3. (a) What are the various parameters of Chromatography ? (CO3)

P. T. O.

(b) Explain Electrochromatography. What are the advantages of Capillary Zone electrophoresis ? (CO3)

(c) Write a short note on SDS PAGE. Explain composition of SDS PAGE sample buffer. (CO3)

4. (a) Write short notes on Ultrafiltration and reverse osmosis. (CO4)

(b) What is Supercritical fluid ? Discuss its advantages and applications. (CO4)

(c) Describe aqueous two phase extraction. What is advantage of this method over solvent extraction ? (CO4)

5. (a) What are various strategies for waste water treatment adopted in Industry ? (CO5)

(b) Explain Microscale bioprocessing utilized in Bioseparation industry. (CO5)

(c) Describe different types of pollutants generated in pharmaceutical Industry. (CO5)